GeoSR-AI: A Deep Learning Framework for 4x Super-Resolution Mapping and Downstream Geospatial Analytics on Sentinel-2 Imagery

Abstract— Open-access optical earth observations from Sentinel-2 provide 10 m Ground Sampling Distance (GSD) across visible and near-infrared (RGBN) bands. However, localized precision agriculture, urban infrastructure mapping, and emergency flood delineation require sub-4 m spatial resolution. This paper presents GeoSR-AI, a deep learning super-resolution mapping framework that transforms 10 m Bottom-of-Atmosphere (BOA) Sentinel-2 imagery into calibrated ~2.5 m spatial products (4x enhancement). To eliminate frequency-domain memory collisions in SEN2SRLite, we design an invariant 128x128 LR chunking and edge-replicate padding protocol coupled with an 8-pass dihedral Test-Time Augmentation (TTA), Radiometric Alignment (RAD), and Iterative Back-Projection (IBP) pipeline. Self-consistency validation achieves PSNR = 38.4186 dB, SSIM = 0.9407, SAM = 0.0223 rad, and RMSE = 0.0120. Downstream modules provide 2.5 m NDVI, 2.5 m NDWI, native 10 m NDBI, and automated flood extent extraction isolating 0.2212 km² of inundated terrain deployed via a React/FastAPI architecture.

Keywords— *Super-Resolution Mapping, Sentinel-2, Deep Learning, Remote Sensing, Spectral Consistency, Change Detection, Flood Assessment.*

I. INTRODUCTION

Multispectral satellite imagery from ESA's Sentinel-2 constellation operates on a 5-day revisit cycle at 10 m GSD for visible/NIR bands (B02, B03, B04, B08) and 20 m for SWIR channels (B11, B12). While suited for regional monitoring, 10 m resolution induces mixed-pixel averaging over small crop parcels, fine road networks, and urban drainage. Sub-meter commercial satellite tasking offers high spatial clarity but remains cost-prohibitive for continuous monitoring.

Deep learning-based Single-Image Super-Resolution (SISR) addresses this gap by synthesizing high-frequency details. However, unconstrained neural models risk 'AI hallucinations'—generating unphysical artifacts and spectral distortions that corrupt biophysical indices. GeoSR-AI provides a physics-constrained 4x super-resolution and analytics platform for Problem Statement 26142, ensuring spectral preservation, memory stability, and error transparency.

II. SYSTEM ARCHITECTURE & METHODOLOGY

A. Invariant 128x128 Chunking Protocol

The SEN2SRLite model incorporates Fast Fourier Transform (FFT) frequency-domain constraints cached to the spatial dimensions of the initial input tensor (128x128 LR -> 512x512 SR). Arbitrary input bounding boxes (e.g., 512x512 LR) trigger PyTorch dimension mismatch errors. GeoSR-AI enforces an invariant tiling protocol:

- **1) Replicate Padding:** Given input (C, h, w), pad height and width: p_h = (128 - h mod 128) mod 128, p_w = (128 - w mod 128) mod 128 using edge replication.
- **2) Uniform Chunk Execution:** Extract non-overlapping 128x128 LR blocks, execute optimized 4x inference, and stitch sub-tiles into a (C, 4H_pad, 4W_pad) canvas.
- **3) Exact Crop:** Crop output to exact target bounds (C, 4h, 4w), eliminating boundary artifacts and guaranteeing stable GPU VRAM overhead.

B. Multi-Stage Optimization Pipeline

- **Dihedral 8-Pass TTA:** Evaluates 8 discrete rotations and horizontal flips per chunk: I_TTA = 1/8 * sum(D_k^(-1)(f_theta(D_k(I_LR)))), removing directional artifacts.
- **Radiometric Alignment (RAD):** Enforces energy conservation via affine gain-bias matching against low-resolution observations: Gain_b = clamp(sigma(I_LR,b) / (sigma(I_hat_LR,b) + eps), 0.5, 2.0).
- **Iterative Back-Projection (IBP):** Applies 3 iterations of residual error correction: I_SR^(t+1) = I_SR^(t) + Bicubic(I_LR - AreaDownsample(I_SR^(t))).

III. GEOSPATIAL ANALYTICS & DISASTER ASSESSMENT

Multi-spectral indices are derived at 2.5 m resolution with an epsilon = 1e-6 safeguard:

- **2.5 m NDVI (Vegetation):** NDVI = (B08 - B04) / (B08 + B04 + eps). Threshold > 0.30 identifies dense canopy; 0.10 < NDVI <= 0.30 isolates crop stress.
- **2.5 m NDWI (Water):** NDWI = (B03 - B08) / (B03 + B08 + eps). Values > 0.00 delineate open surface water bodies.
- **Native 10 m NDBI (Urban):** NDBI = (B11 - B08) / (B11 + B08 + eps). Computed with native SWIR (B11) to isolate built-up infrastructure (NDBI > 0.00).
- **Temporal Flood Inundation (Delta-NDWI):** Delta_NDWI = NDWI_after - NDWI_before. Flooded Area (km²) = sum(Delta_NDWI > 0.20) * (2.5 / 1000)^2.

IV. EXPERIMENTAL RESULTS & BENCHMARKS

Evaluation was conducted on Sentinel-2 Level-2A data over Vijayawada, India (16.506667 N, 80.629468 E; Scene Timestamp: 2023-12-31T05:02:19Z; Valid Pixels: 99.6%; 512x512 LR tile). Self-consistency degradation testing benchmarked the engine against native observations.

Pipeline Configuration	PSNR (Area)	SSIM (Area)	PSNR (Bic)	SSIM (Bic)
Bicubic Baseline	38.0760	0.9361	38.0603	0.9352
SR Plain	37.6928	0.9344	38.2261	0.9396
SR + TTA	38.0030	0.9367	38.5405	0.9417
SR + TTA + RAD	38.0022	0.9367	38.5327	0.9417
SR + TTA + RAD + IBP	**38.4186**	**0.9407**	**38.4576**	**0.9412**

Metric	GeoSR-AI (SEN2SR Improved)	Bicubic Baseline
PSNR (dB)	**38.4186 dB**	38.0760 dB
SSIM	**0.9407**	0.9361
SAM (rad)	**0.0223 rad**	0.0224 rad
RMSE / MAE	**0.0120 / 0.0076**	0.0125 / 0.0079
Reconstruction Consistency Error	**0.0003**	N/A

- **Spectral Bands:** B04 (Red): RMSE=0.0120 | B03 (Green): RMSE=0.0107 | B02 (Blue): RMSE=0.0108 | B08 (NIR): RMSE=0.0141.
- **Land Cover Extents:** Vegetation (NDVI>0.3): 6.85% | Crop Stress (0.1<NDVI<=0.3): 31.46% | Water (NDWI>0): 21.86% | Built-Up (NDBI>0): 35.84%.
- **Flood Delineation (Feb vs. Aug 2023):** Extracted 0.2212 km² (3.38% of AOI) of inundated surface terrain at Delta-NDWI > 0.20.

V. EXPLAINABILITY, DEPLOYMENT & LIMITATIONS

GeoSR-AI enforces operational transparency by generating pixel-wise spatial RMSE error heatmaps and inverse-enhancement confidence proxies (Confidence = 1.0 - Norm(|SR - Bicubic|)), clearly isolating model-synthesized textures from sensor observations.

The stack is deployed via an asynchronous FastAPI backend and a React 19 / MapLibre GL frontend with dynamic coordinate picking and base64 raster streaming.

Limitations: Evaluation relies on downscaled self-consistency reference targets rather than aerial sensors. Single-threshold Delta-NDWI remains sensitive to cloud shadows.

VI. CONCLUSION

GeoSR-AI resolves FFT dimensional memory collisions, delivers 4x spatial super-resolution mapping (PSNR = 38.4186 dB, SSIM = 0.9407), and provides automated multi-spectral analytics and flood disaster mapping for operational remote sensing.

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